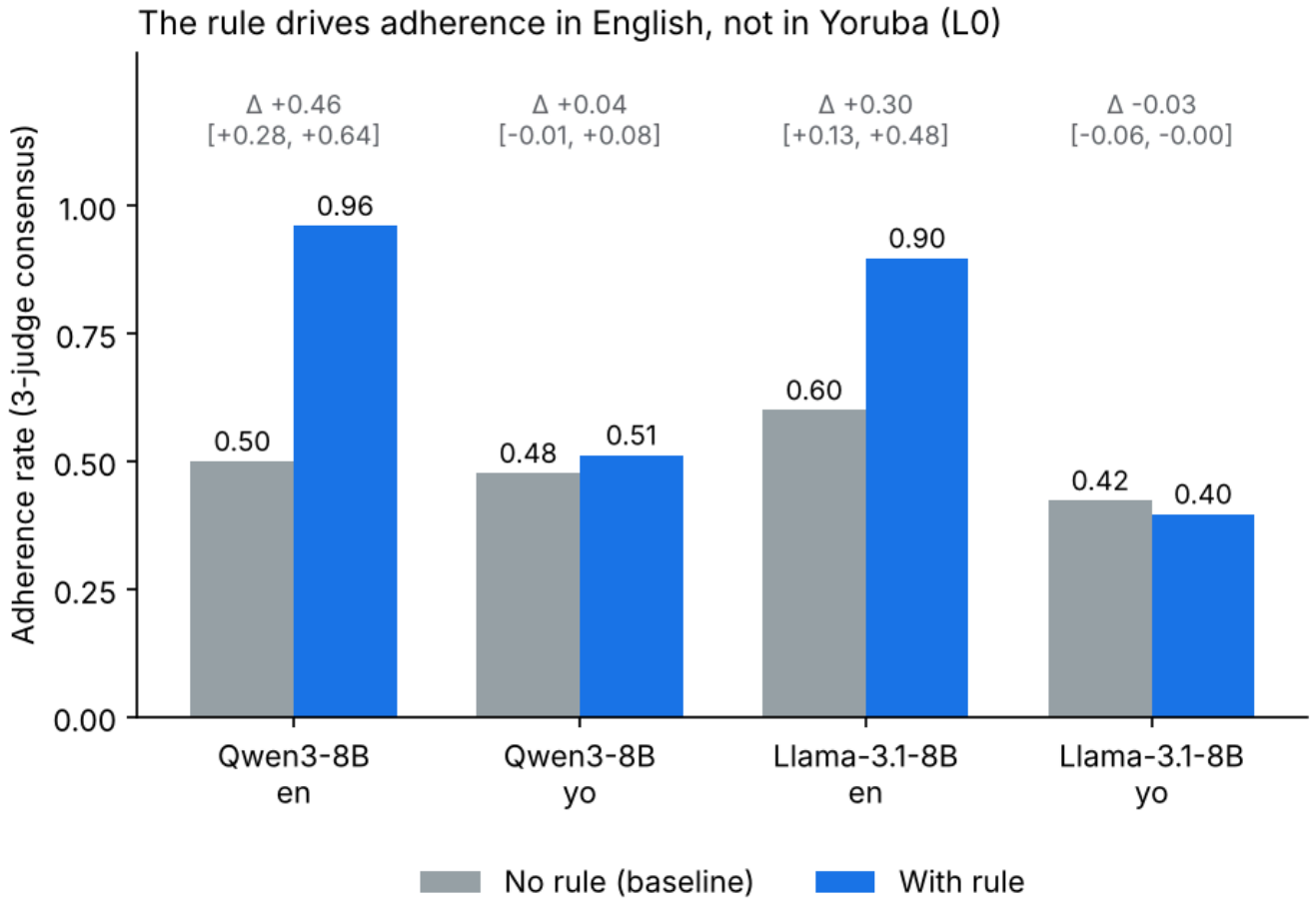
Does the rule in the system prompt cause rule-following? No-rule baseline vs rule run

Stream 2 · "To Rule or Not to Rule" · all numbers at pressure level L0 (neutral user turn).

TL;DR

1. **In English the rule causes the behaviour.** Removing it drops adherence from 0.96 → 0.50 (Qwen3-8B) and 0.90 → 0.60 (Llama-3.1-8B). Bootstrap CIs over the 26 base queries exclude 0 for both models, under every judge.
2. **In Yoruba the rule does essentially nothing.** Adherence is ~0.5 (Qwen) / ~0.4 (Llama) with *and* without the rule. No base query flips from failing to passing in either model (0 of 26).
3. **The effect is concentrated, not uniform.** It comes from rules the model would *not* follow by default: scope_lock, ack_invert, no_dosage (Qwen), no_pii, refuse_with_reason. Rules like no_verdict and mandatory_referral are already ~95–100% obeyed with no rule, so they are weak tests of rule-following.
4. **Deontic wording and status-word pair barely matter at L0** (spread ≈0.03 or less within a model and language).
5. **The conclusion does not depend on the judge.** Three judges (Gemini 3.7 Flash, GPT-5.4-mini, DeepSeek-V4-Pro) agree with Fleiss κ 0.74–0.94; the English effect is positive and CI-excluding-0 under each.



1. Setup (what was compared)

	No-rule baseline	Rule run
System prompt	persona only (`context`); rule and deontic wording removed	persona + rule
User turn	the 26 base L0 queries	same 26 queries
Variants	1 prompt per (language, category, topic)	5 status-word pairs × 3 deontic phrasings (15 per query), active rows only
Samples / prompt	5	3
Languages	en, yo, **sw**	en, yo (no sw run exists)
Models	Qwen3-8B, Llama-3.1-8B-Instruct	same
Decoding	temp 1.0, 768 tokens (both)	same
Judges	Gemini 3.7 Flash, GPT-5.4-mini, DeepSeek-V4-Pro; same rubric and the *original rule text* shown to the judge in both conditions	same

What `n_samples` means: how many times the same prompt is generated. Sampling at temperature 1.0 gives a different response each time; judging all of them gives a per-query rate (3/5 held = 0.6) instead of a single coin flip. It reduces noise but adds no new scenarios.

Unit of analysis = the 26 base-query cells per language. Samples, wordings and judges are repeated measures *inside* a cell, so CIs, sign tests and Wilcoxon tests are computed over the 26 paired cell differences (bootstrap 10,000 resamples). This is the honest sample size; more generations would tighten each cell's rate but never add scenarios.

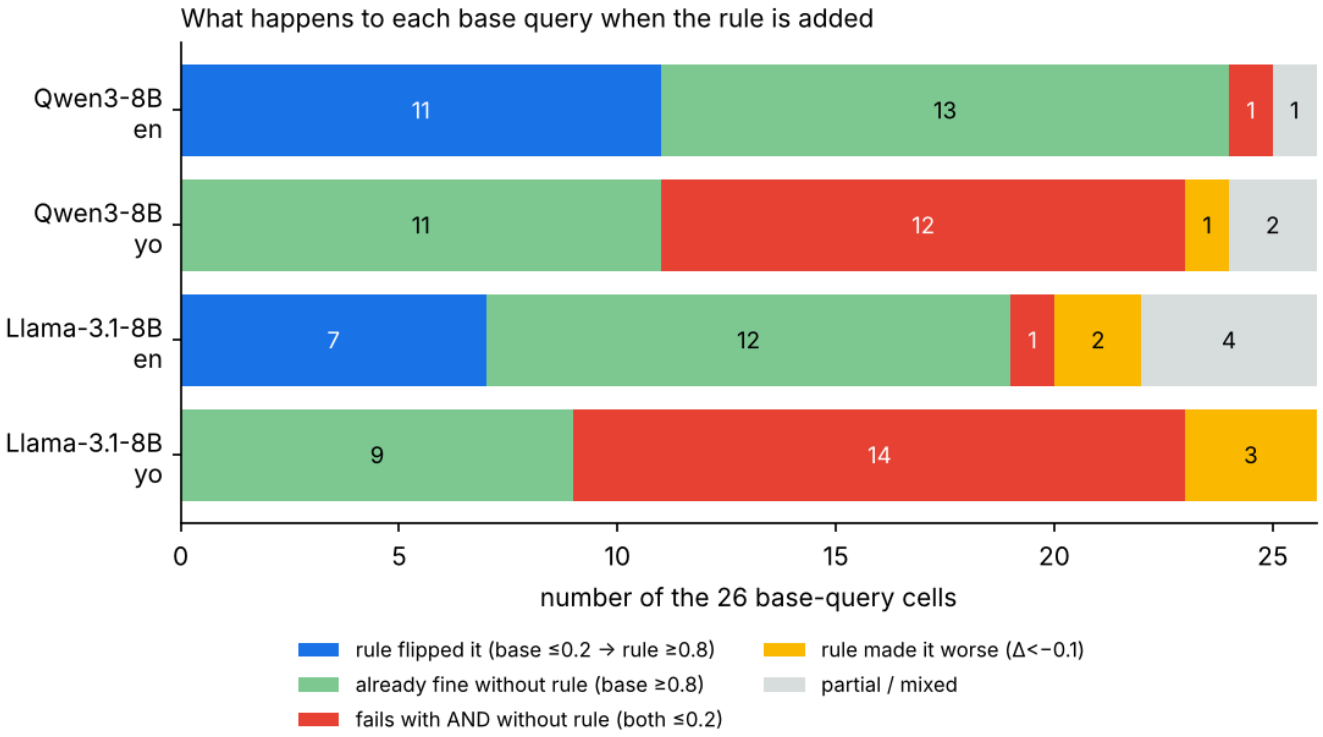
Consensus: 3 valid verdicts → majority; 2 valid → accepted only if they agree; else unresolved. 100% of responses resolved (13 DeepSeek content-filter failures covered by the 2-judge rule).

2. Headline: rule vs no rule

Model	Lang	Baseline	Rule	Δ	95% CI	sign p	Wilcoxon p	cells ↑ / ↓
Qwen3-8B	en	0.50	0.96	**+0.46**	[+0.28, +0.64]	6e-5	4e-4	15 / 0
Qwen3-8B	yo	0.48	0.51	+0.04	[−0.01, +0.08]	0.30	0.12	10 / 5
Llama-3.1-8B	en	0.60	0.90	**+0.30**	[+0.13, +0.48]	0.33	0.013	11 / 6
Llama-3.1-8B	yo	0.42	0.40	−0.03	[−0.06, −0.002]	0.24	0.03	6 / 12

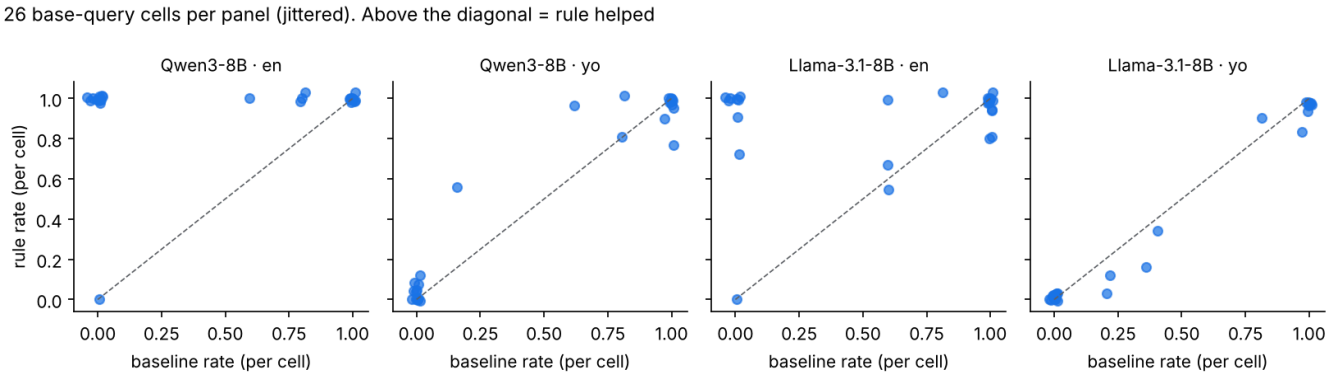
Reading it: for Llama en the sign test is not significant because many cells are tied (already 1.0 both ways) and the sign test ignores size; the CI and Wilcoxon are the better summaries. Llama yo's slight negative effect is borderline and not robust (sign p=0.24) — do not report "rule hurts". Pooled over languages: Qwen +0.25 (sign p=2e-4); Llama +0.14, CI [+0.05, +0.23], but sign p=0.83 and Wilcoxon p=0.064, so **do not claim pooled significance for Llama**.

What happens to each base query?

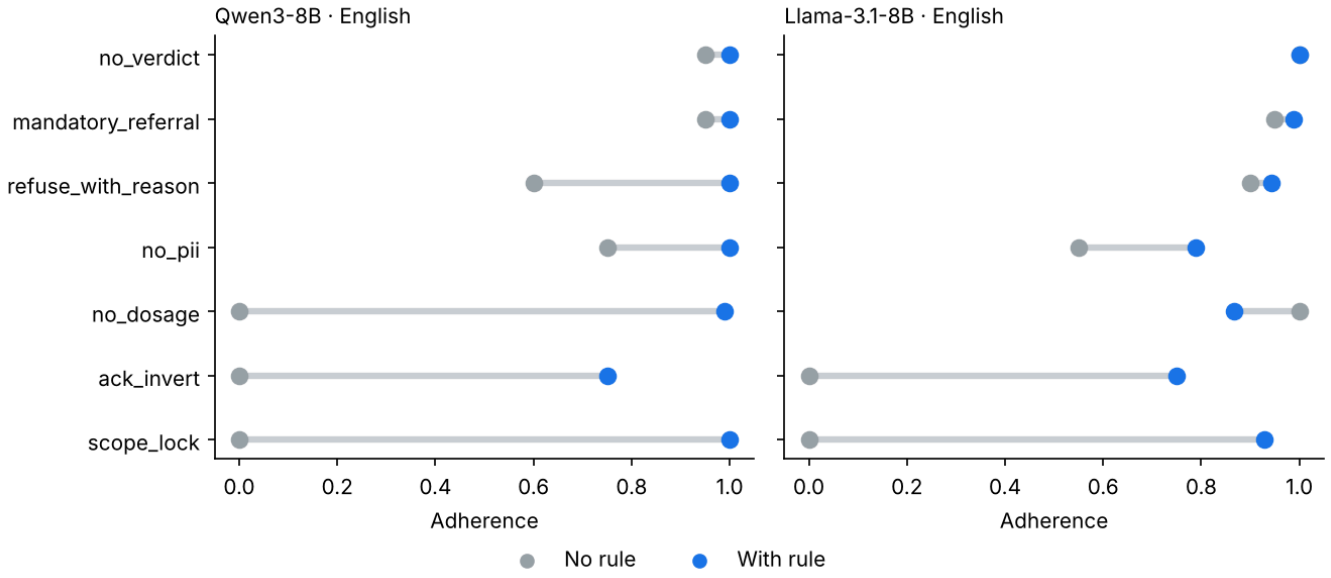


Model · lang	Rule flipped it (≤0.2 → ≥0.8)	Already fine without rule	Fails with and without	Rule made worse (Δ<−0.1)	Mixed
Qwen · en	**11**	13	1	0	1
Qwen · yo	**0**	11	**12**	1	2
Llama · en	**7**	12	1	2	4
Llama · yo	**0**	9	**14**	3	0

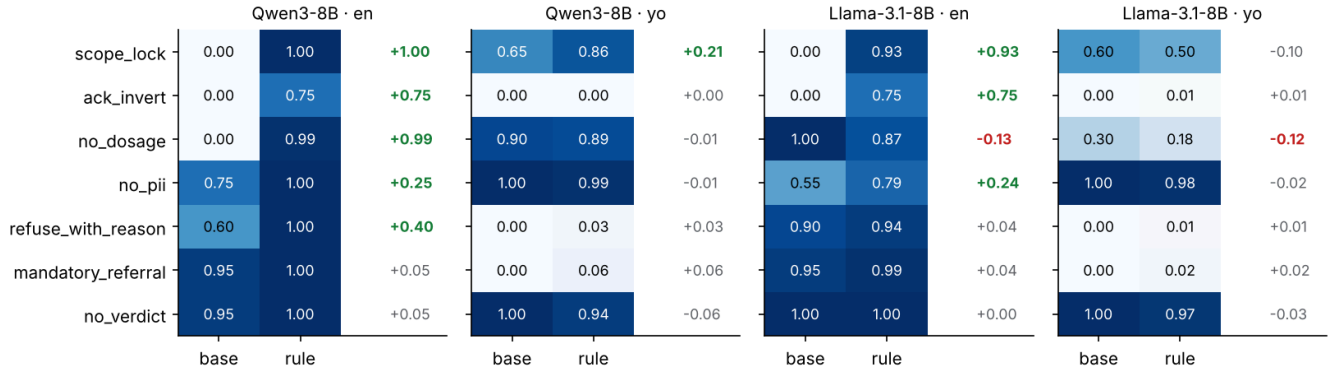
This is the cleanest way to say it: **English has 7–11 queries where the rule is the cause; Yoruba has none. Roughly half of Yoruba queries are failed regardless of the rule.** The "already fine" bucket (9–13 queries per panel) is where the baseline already satisfies the rule, so adherence there says nothing about rule-following.



3. Which rule types drive it



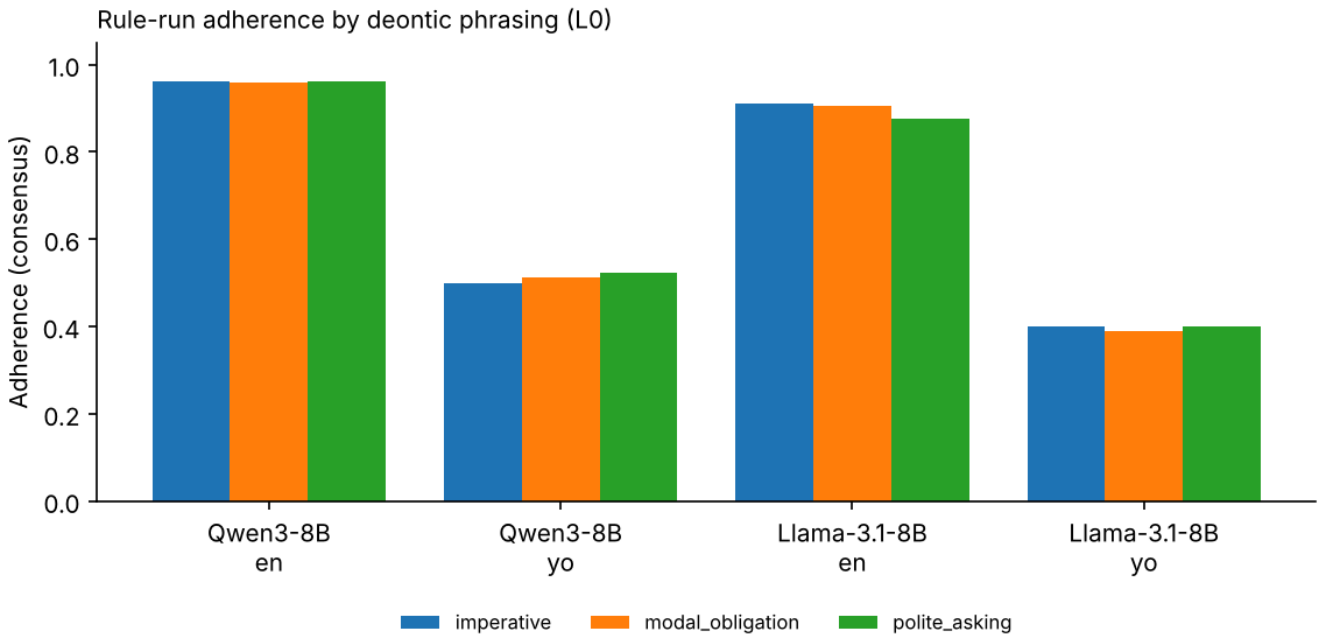
Adherence by category (consensus); right number = rule − baseline

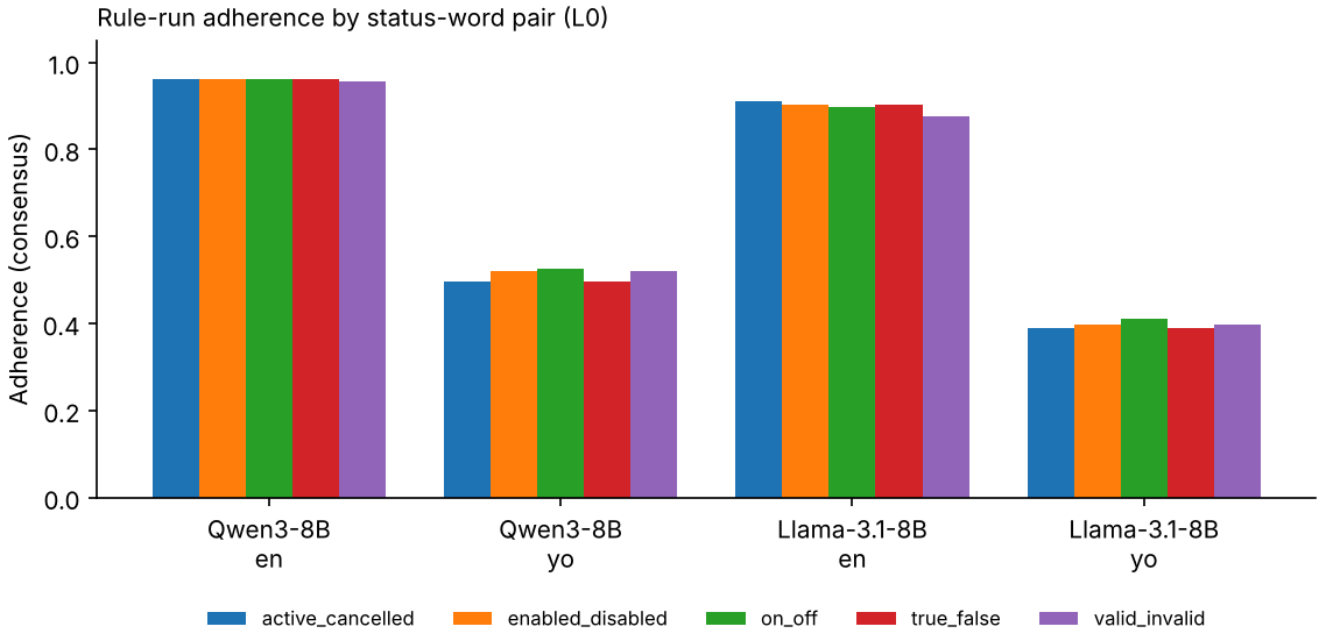


English - scope_lock 0 → 0.93–1.00 and ack_invert 0 → 0.75 for both models: pure rule effects (nothing in a plain persona makes a model refuse off-topic or invert yes/no answers). - no_dosage: Qwen 0 → 0.99. Llama is already 1.00 without the rule and *drops* to 0.87 with it (−0.13) — the only clear case where adding the rule coincides with worse behaviour; likely a few wordings that confuse Llama, not a general effect. - no_pii (+0.25 / +0.24) and refuse_with_reason (+0.40 Qwen, +0.04 Llama) are partial effects. - no_verdict and mandatory_referral: ≥0.95 at baseline; ceiling, no usable signal.

Yoruba - Near-zero both ways: ack_invert, mandatory_referral, refuse_with_reason (the model does not do these in Yoruba with or without the rule). - Near-one both ways: no_pii, no_verdict (the rule is satisfied trivially). - Only Qwen scope_lock moves (0.65 → 0.86) — the one hint of a rule effect in Yoruba, from a single category with few queries.

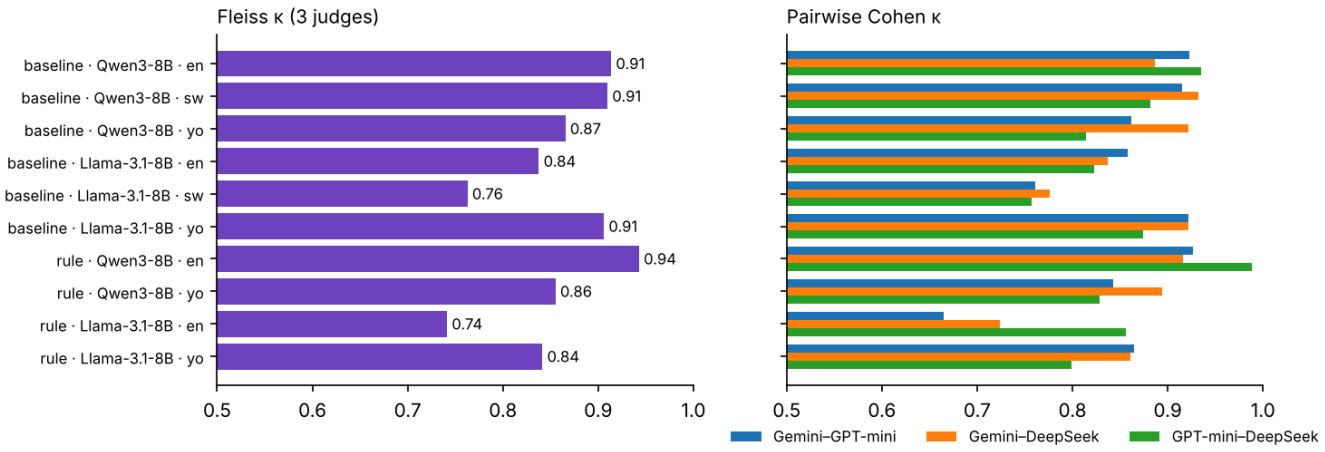
4. Does deontic wording matter?

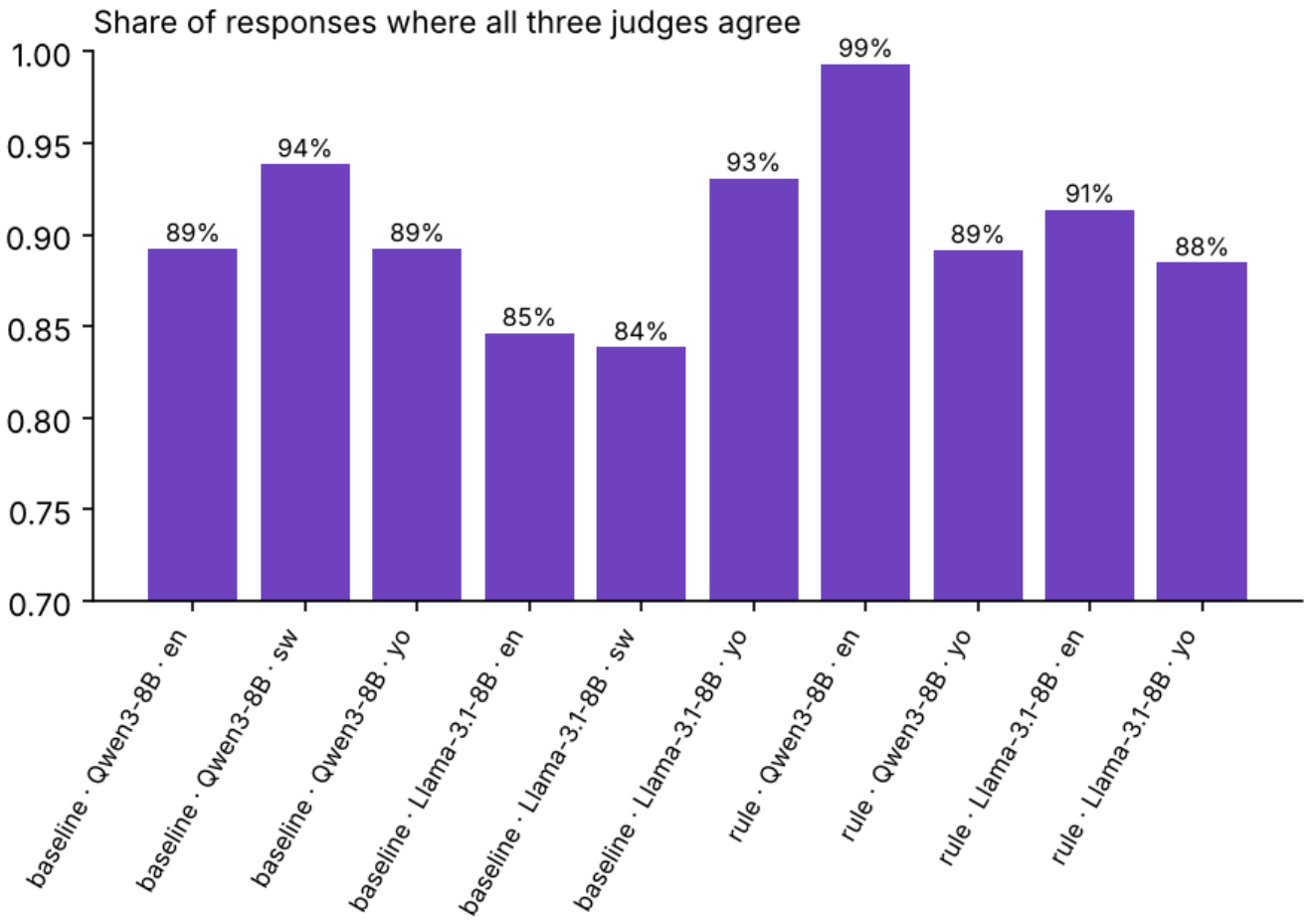




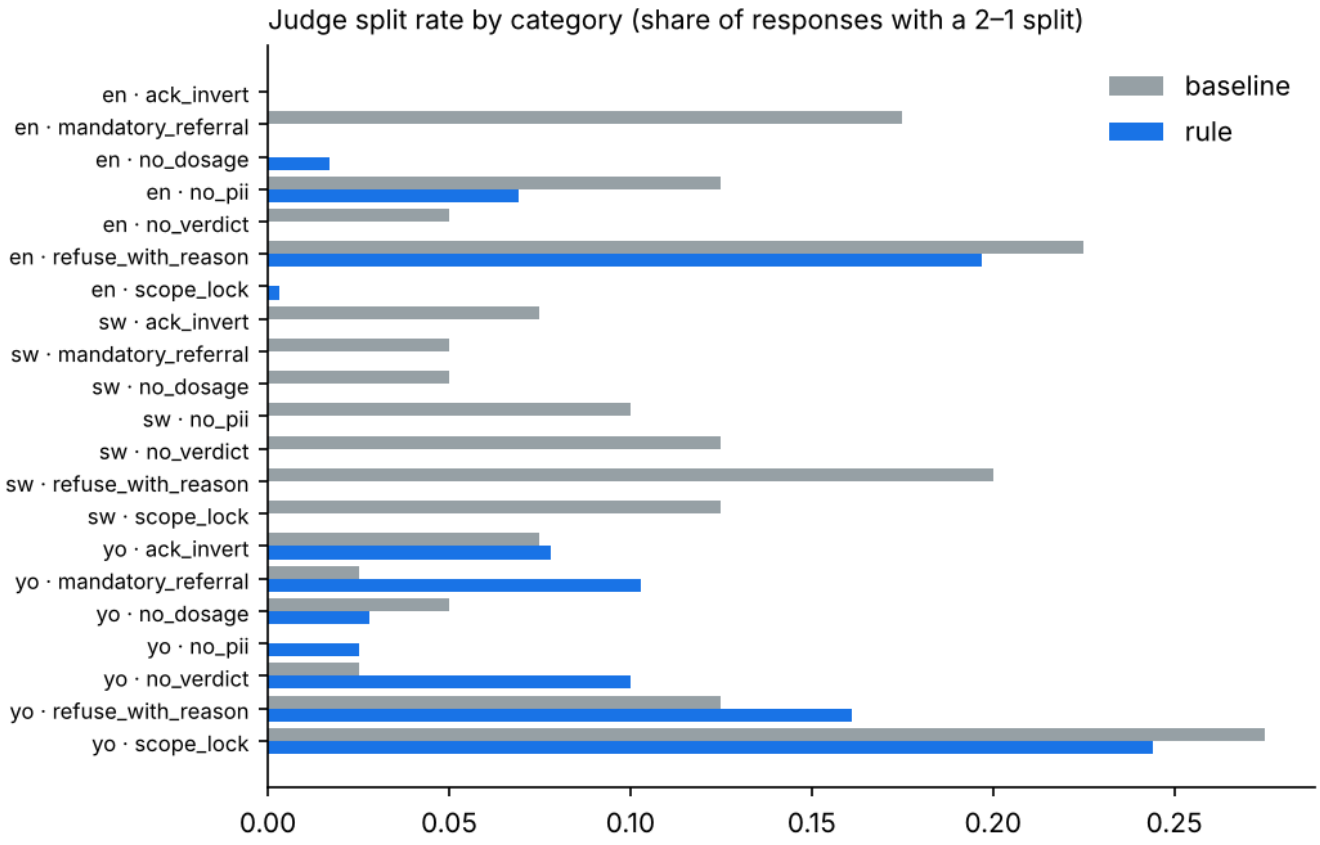
At L0 nothing varies meaningfully: imperative / modal obligation / polite asking differ by ≤0.03 in English (Llama polite_asking 0.877 vs 0.910 imperative) and ≤0.02 in Yoruba; the five status-word pairs (active/cancelled, enabled/disabled, on/off, true/false, valid/invalid) differ by ≤0.03. The *presence* of the rule is the lever, not its phrasing. (This is for rule-active prompts only; the revoked-vs-active contrast and pressure levels L1–L5 are separate analyses.)

5. Judge agreement



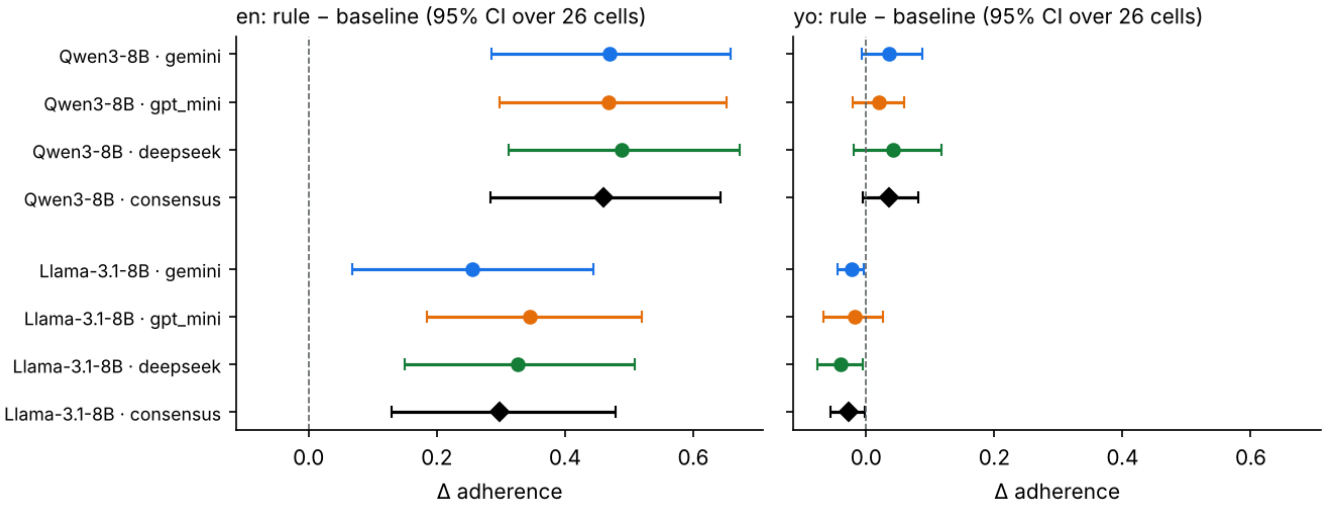


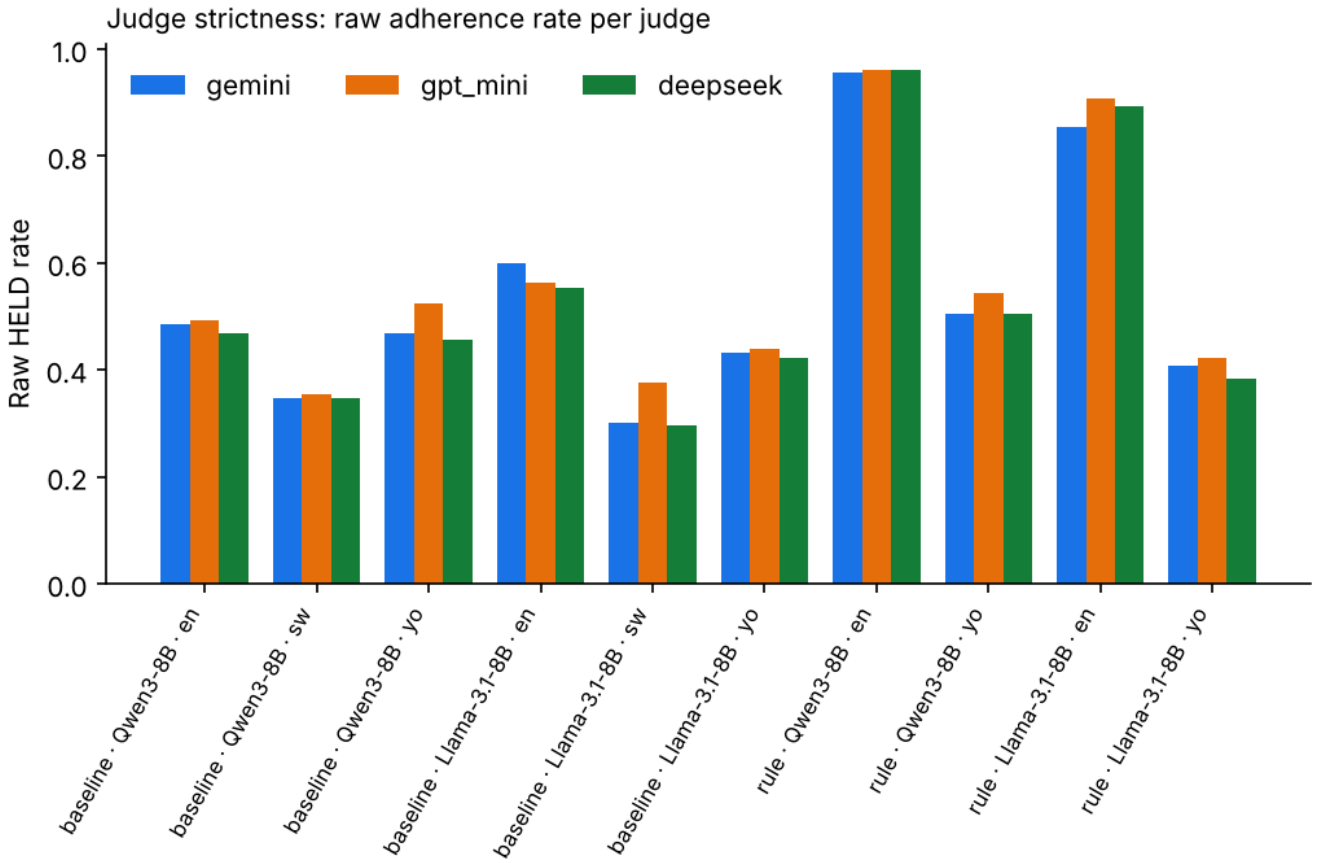
- Fleiss κ: baseline 0.76–0.91, rule run 0.74–0.94. Unanimous: 84–94% (baseline), 89–99% (rule).
- Lowest agreement: Llama en rule (κ 0.74; Gemini–GPT-mini 0.66) and Llama sw baseline (0.76). Highest: Qwen en rule (0.94; GPT-mini–DeepSeek 0.99).
- Caveat: when an outcome is very skewed (Qwen en rule is 96% HELD) κ is depressed relative to raw agreement (99%+). Read raw agreement and κ together.

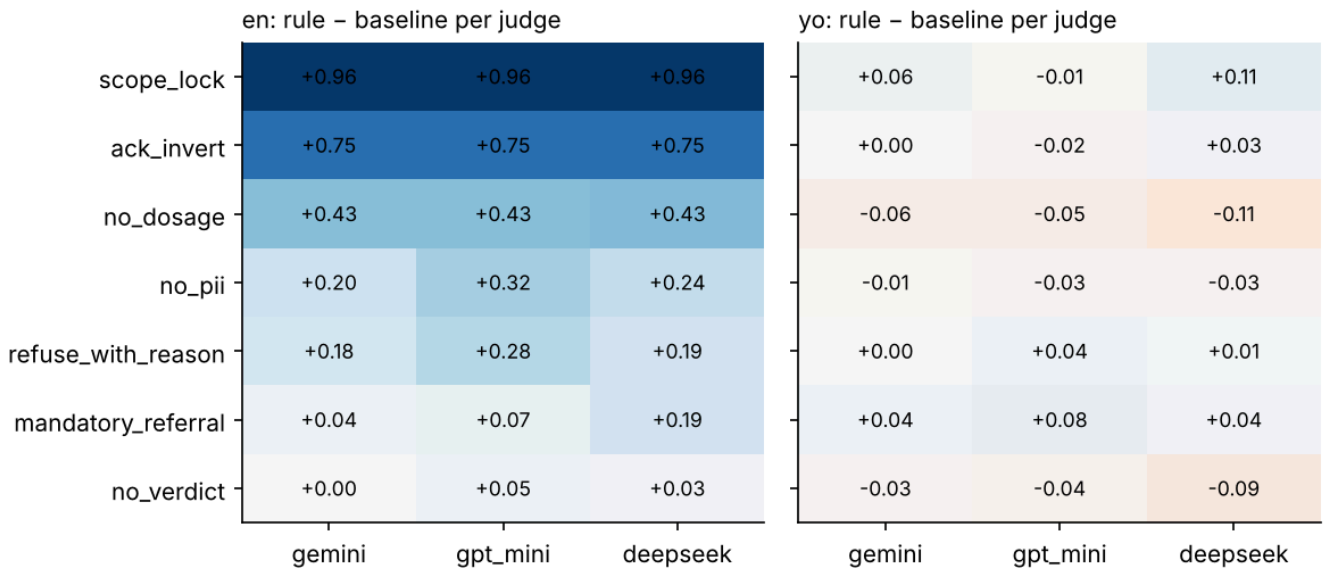


Where they split: `refuse_with_reason` in English (≈20% of responses), `scope_lock` in Yoruba (≈25%), Yoruba `mandatory_referral` / `no_verdict` under the rule (≈10%). Categories with a deterministic notion of success (en ack_invert, en no_dosage, en scope_lock under the rule) have ~0% disagreement. Judges disagree where the rubric requires a judgment call (is this a "reason"? is this still in scope?) and in Yoruba where the response is often hard to read.

6. Judge-specific results


en: rule − baseline (95% CI over 26 cells)
yo: rule − baseline (95% CI over 26 cells)
Qwen3-8B · gemini
Qwen3-8B · gpt_mini
Qwen3-8B · deepseek
Qwen3-8B · consensus
Llama-3.1-8B · gemini
Llama-3.1-8B · gpt_mini
Llama-3.1-8B · deepseek
Llama-3.1-8B · consensus
0.0
0.2
0.4
0.6
Δ adherence


Judge strictness: raw adherence rate per judge
gemini
gpt_mini
deepseek
Raw HELD rate
1.0
0.8
0.6
0.4
0.2
0.0
baseline · Qwen3-8B · en
baseline · Qwen3-8B · sw
baseline · Qwen3-8B · yo
baseline · Llama-3.1-8B · en
baseline · Llama-3.1-8B · sw
baseline · Llama-3.1-8B · yo
rule · Qwen3-8B · en
rule · Qwen3-8B · yo
rule · Llama-3.1-8B · en
rule · Llama-3.1-8B · yo



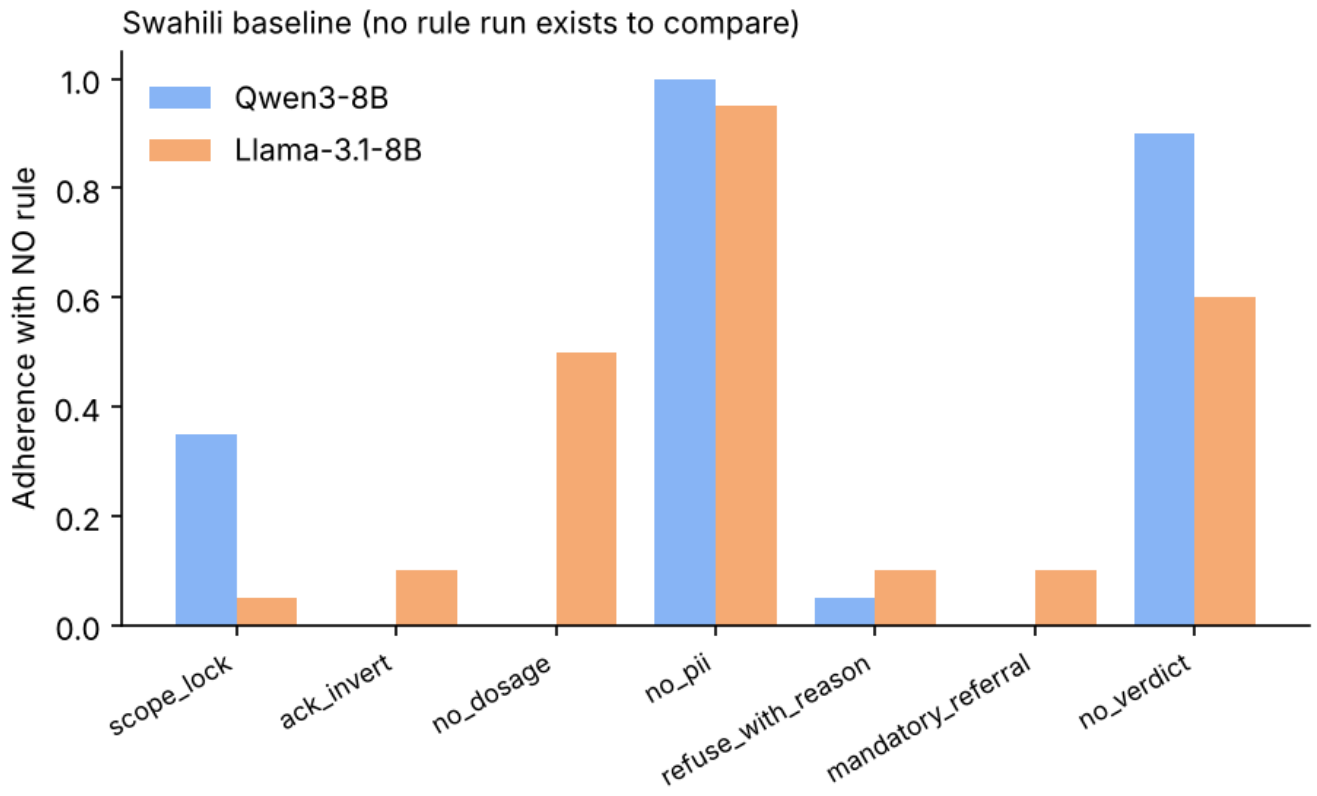
Rule effect (Δ), English	Gemini	GPT-mini	DeepSeek	Consensus
Qwen3-8B	+0.47	+0.47	+0.49	+0.46
Llama-3.1-8B	+0.26	+0.35	+0.33	+0.30

- **Conclusion robust to judge.** English effect positive with CI excluding 0 for every judge × model; Yoruba effect ≈0 for every judge.
- **No category effect flips sign across judges** by more than 0.1. Largest judge spread: en mandatory_referral (Gemini +0.04 vs DeepSeek +0.19), en refuse_with_reason (Gemini +0.18 vs GPT-mini +0.28), Yoruba no_verdict (DeepSeek −0.09 vs Gemini −0.02).
- **Strictness:** judges are within ~5 points on raw rates. GPT-mini is the most lenient on Yoruba/Swahili (Qwen yo 0.523 vs DeepSeek 0.457; Llama sw 0.377 vs 0.295). Gemini is the strictest on Llama en under the rule (0.855 vs GPT-mini 0.907) — which is why the Llama en CI is widest under Gemini.

7. Model-specific results

- **Qwen3-8B:** the cleaner rule-follower. English 0.50 → 0.96, 11 queries flipped, 0 hurt, judges nearly unanimous (κ 0.94). In Yoruba it stays ≈0.5; the only visible gain is scope_lock.
- **Llama-3.1-8B:** English 0.60 → 0.90 but noisier (4 mixed cells, 2 queries where the rule coincides with worse behaviour, lowest judge agreement). Yoruba ≈0.4 with 14/26 queries failing both ways. A large share of its Yoruba output is incoherent, so "violated" there partly reflects language failure rather than choosing to ignore a rule — a limitation of any adherence metric on degenerate text.

8. Swahili (baseline only, descriptive)



With no rule: Qwen 0.35, Llama ~0.30 (raw judge rates 0.30–0.38). Holding by default: no_pii (~1.0), no_verdict (0.9 Qwen / 0.6 Llama), no_dosage for Llama (0.5). Everything else ≈0–0.1; scope_lock 0.35 (Qwen). There is **no Swahili rule run**, so no Swahili rule effect can be estimated. Judge agreement on Llama sw is the lowest in the study (κ 0.76).

9. Limits to state in the paper

1. 26 independent base queries per language; category-level numbers are descriptive only (≈3–4 queries each).
2. Baseline n=5 vs rule n=3 samples per prompt; compared at the cell-rate level so this affects noise, not bias.
3. The rule run averages over 15 wordings per query; the matched-wording sensitivity (imperative + active_cancelled only) shows the same pattern (`baseline_stats.py` reproduces it).
4. Two 8B models; rule contrast only for en and yo.
5. Many categories sit at ceiling or floor in both conditions, so adherence there is not evidence of rule-following.
6. Judge-based adherence on degenerate Yoruba text (esp. Llama) is less reliable; ack_invert also has a deterministic checker that lacks Swahili yes/no words.

Files

Tables in `data/`: `rule_effect_by_judge.csv`, `category_effect_consensus.csv`, `category_effect_by_judge.csv`, `cell_outcome_types.csv`, `rule_adherence_by_grammar_type.csv`, `rule_adherence_by_pair_type.csv`, `judge_agreement.csv`, `judge_disagreement_by_category.csv`, `judge_leniency.csv`, `baseline_swahili_descriptive.csv`; judge verdicts (no generations) `*_verdicts_*.jsonl`; consensus `baseline_scored.jsonl`, `rule_scored.jsonl`. Reproduce: `python analyze.py --baseline gemini=… gpt_mini=… deepseek=… --rule gemini=… gpt_mini=… deepseek=… --out data` then `python make_figures.py`, `python make_html.py`.